Traceable Predictive Maintenance for NASA C-MAPSS

4-minute overview of the deployed PHM application

Benchmark: NASA C-MAPSS turbofan degradation

Goal: turn RUL prediction into usable PHM decisions

Focus: deployment, traceability, and system integration

CMAPSS
PHM App

April 2026

Problem

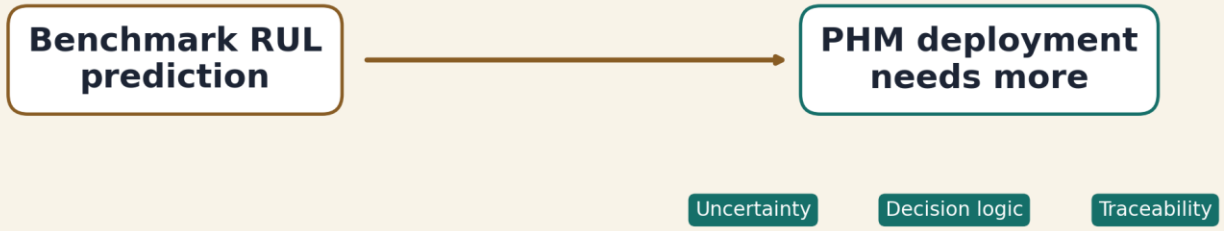
Why benchmark prediction alone is not enough

Benchmark RUL models produce point predictions.

Real PHM systems also need uncertainty, decision logic, and traceability.

Prediction alone is NOT sufficient for PHM deployment.

NOT sufficient for PHM deployment



NOT sufficient for PHM deployment

Task Context

NASA C-MAPSS defines the prediction setting

Run-to-failure engine trajectories

Multivariate sensor + operational data

Goal: predict Remaining Useful Life

System Idea

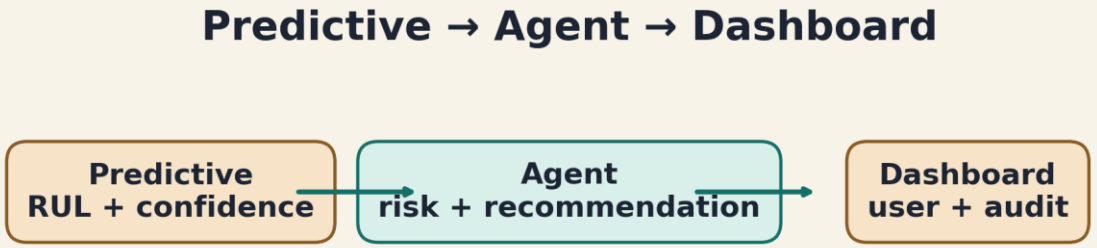
Not just a model, but a system

Predictive → Agent → Dashboard

Predictive: RUL + confidence

Agent: risk + recommendation

Dashboard: user + audit



Predictive → Agent → Dashboard

Predictive Layer

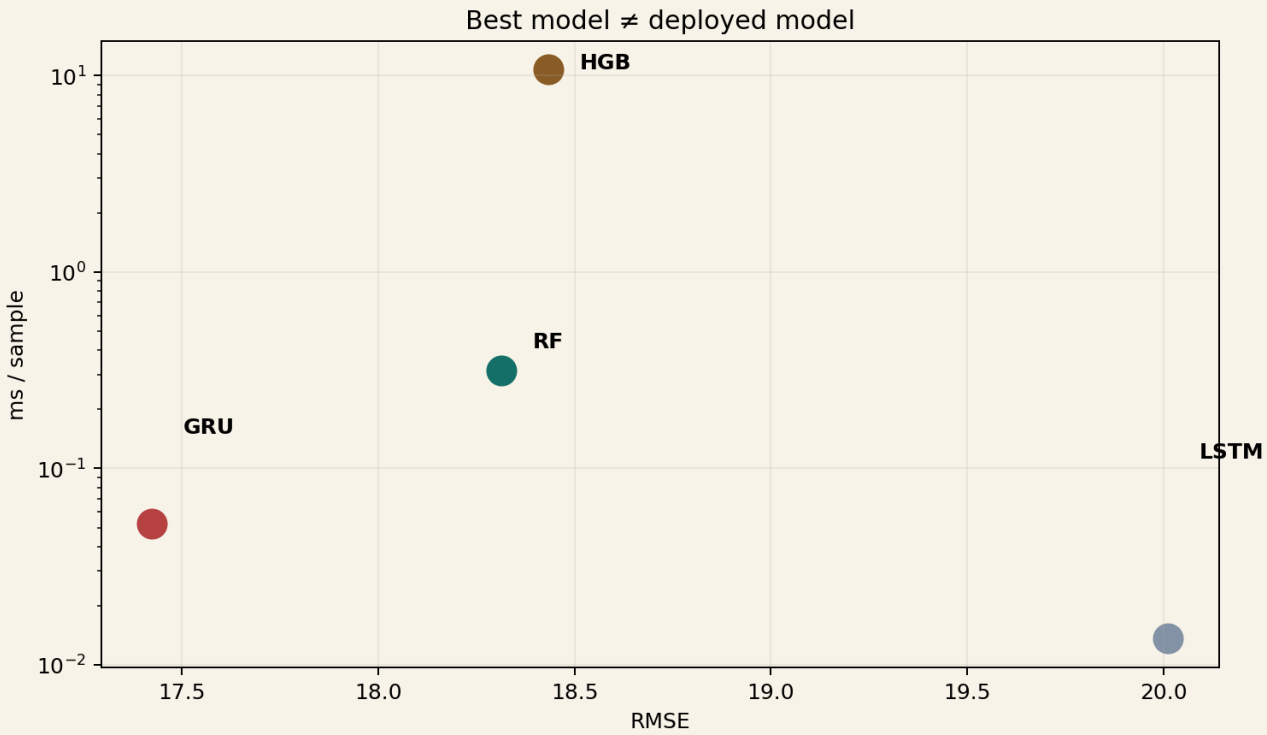
Best model ≠ deployed model

Multiple models tested: RF, HGB, LSTM, GRU

GRU achieved the best raw RMSE

RF was deployed as champion

Deployment needs stability, latency, and robustness



Best model ≠ deployed model

Agent Layer

Prediction becomes decision support

Inputs: RUL, confidence, service status

Outputs: risk level, risk score, recommendation

Deterministic and auditable

LLM optional, never controls decisions

Dashboard

The integrated user-facing system

Summary

→ decision

Analysis

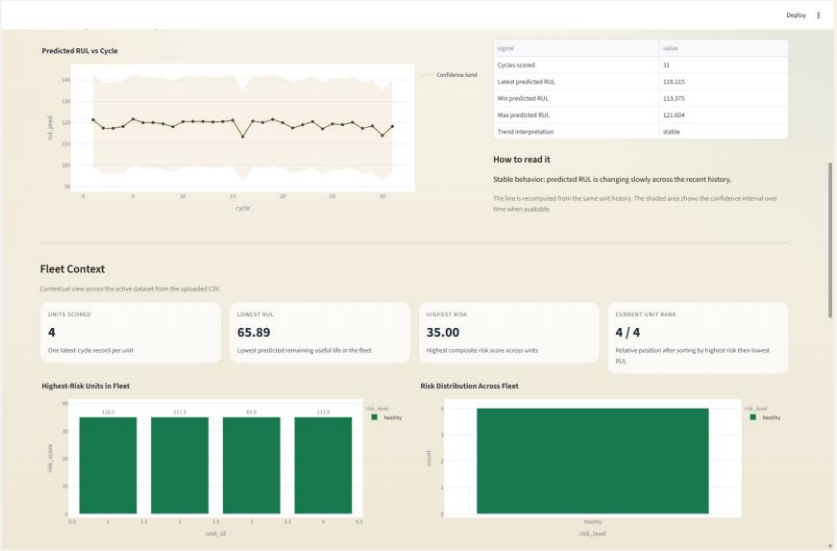
→ uncertainty

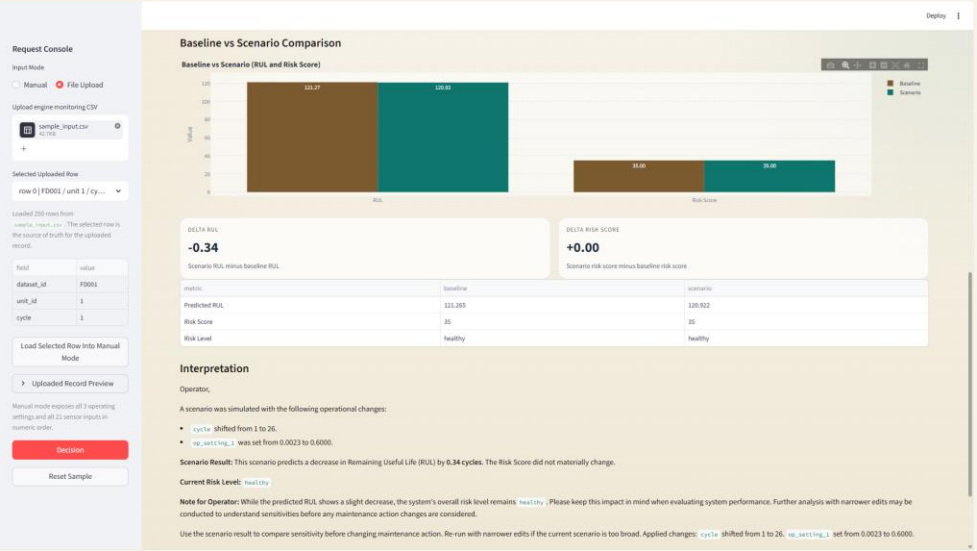
Scenarios

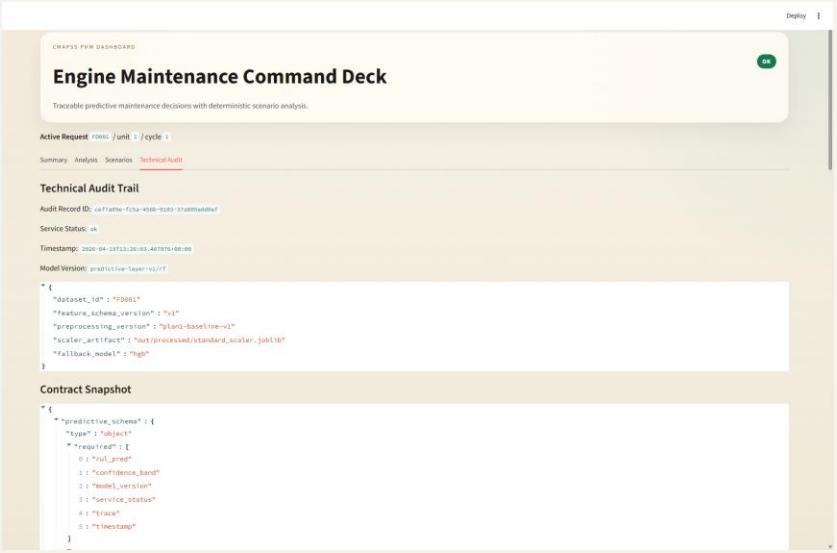
→ what-if comparison

Audit

→ traceability







Results

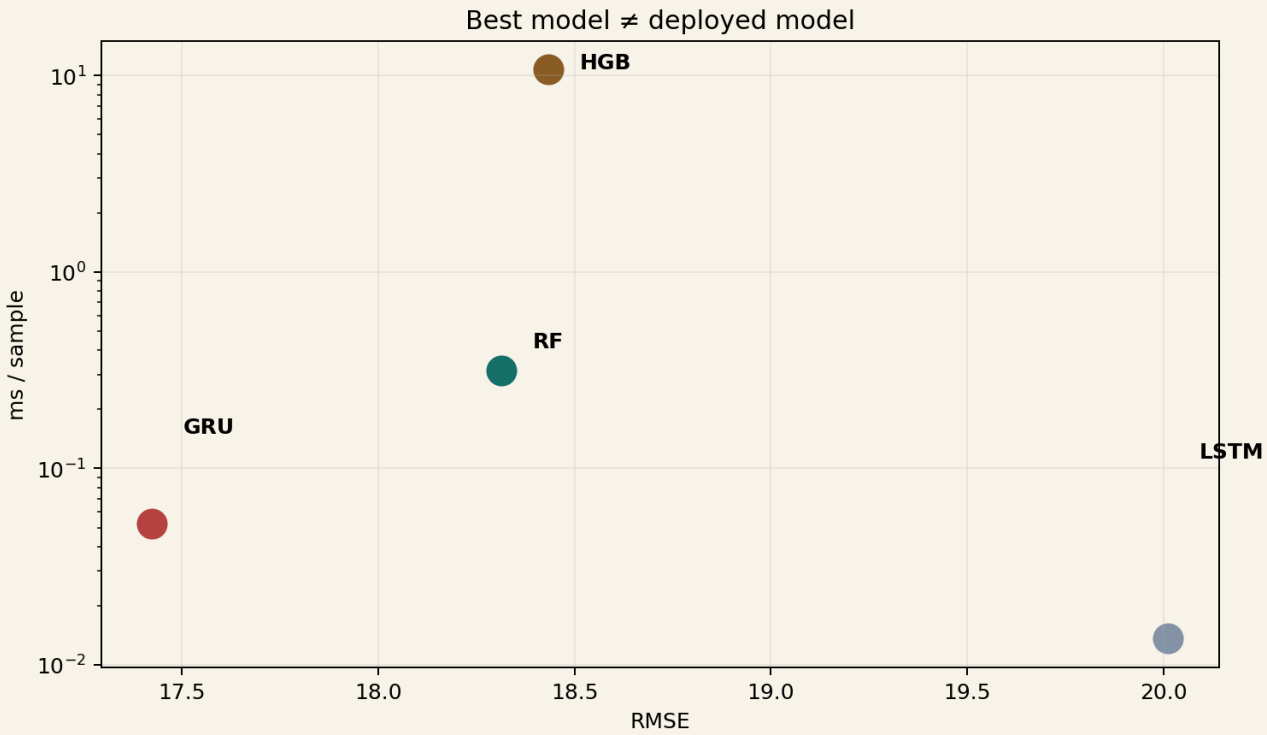
Engineering constraints matter

GRU → best accuracy

RF → best system behavior

Best model ≠ deployed model

PHM quality is not RMSE alone



Best model ≠ deployed model

Scenarios

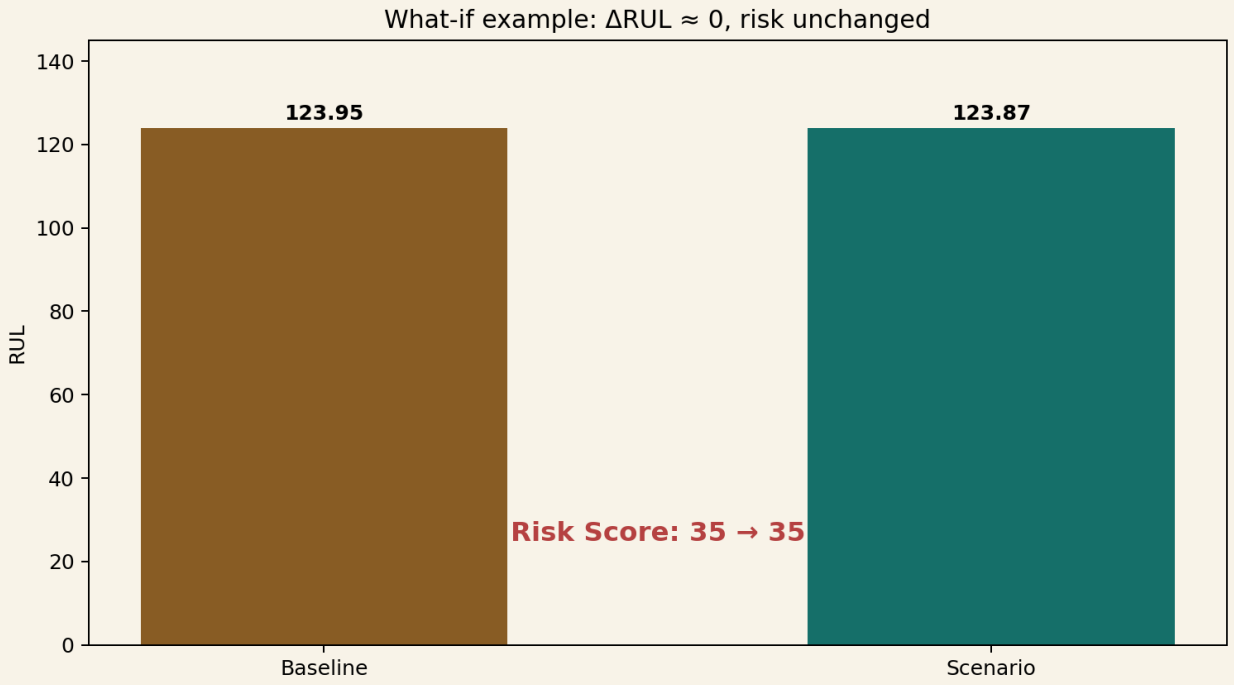
Interpretability through what-if analysis

Example: ΔRUL ≈ 0

Risk unchanged

Not all changes matter

The system reveals sensitivity, not just prediction



Conclusion

A deployable PHM system, not just a model

Calibrated prediction

Deterministic decisions

Traceability and auditability

Main lesson: PHM quality ≠ RMSE alone